

Bayesian Structure of a Model as a Causal Graph

IA pour la Science

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Probabilistic model in deep learning

Def.

The model $\mathbf{f}(\mathbf{w}, \mathbf{x})$ with differentiated parameters $\mathbf{w}$ maps the feature description set to the label set:

$$\mathbf{f} : \mathbb{X} \times \mathbb{W} \rightarrow \mathbb{Y},$$

where $\mathbb{W}$ is space of parameters for $\mathbf{f}$.

The model selection problem challenge: the great number of parameters makes many optimization criteria (like AIC, BIC, CV) inefficient.

The model is defined by its structure $\mathbf{\Gamma}$ and parameters $\mathbf{W}$.

The structure defines superpositions to form the model, which must be selected according to statistical criteria of model complexity.

The empirical estimators of the model complexity could be:

- ① the number of parameters;
- ② number of function in the superposition;
- ③ number of admissible superpositions.

Example: the structure of two-layer perception

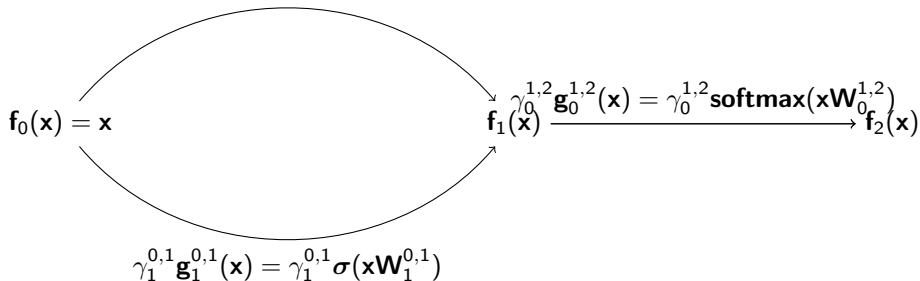
The model $\mathbf{f}$ is defined by the **structure** $\mathbf{\Gamma} = [\gamma^{0,1}, \gamma^{1,2}]$.

$$\text{Модель: } \mathbf{f}(\mathbf{x}) = \text{softmax} \left(\mathbf{f}_1(\mathbf{x}) \mathbf{W}_0^{1,2} \right), \quad \mathbf{f}(\mathbf{x}) : \mathbb{R}^n \rightarrow [0, 1]^{|\mathbb{Y}|}, \quad \mathbf{x} \in \mathbb{R}^n.$$

$$\mathbf{f}_1(\mathbf{x}) = \gamma_0^{0,1} \mathbf{g}_0^{0,1}(\mathbf{x}) + \gamma_1^{0,1} \mathbf{g}_1^{0,1}(\mathbf{x}),$$

where $\mathbf{w} = [\mathbf{W}_0^{0,1}, \mathbf{W}_1^{0,1}, \mathbf{W}_0^{1,2}]^\top$ are the parameters, $\{\mathbf{g}_{0,1}^0, \mathbf{g}_{0,1}^1, \mathbf{g}_{1,2}^0\}$ a the link function for the hidden layers.

$$\gamma_0^{0,1} \mathbf{g}_0^{0,1}(\mathbf{x}) = \gamma_0^{0,1} \sigma(\mathbf{x} \mathbf{W}_0^{0,1})$$



The model structure as a superposition graph

There given:

- ① acyclic graph (V, E) ;
- ② each edge $(j, k) \in E$ corresponds to its vector of basic differentiable functions $\mathbf{g}^{j,k} = [\mathbf{g}_0^{j,k}, \dots, \mathbf{g}_{K^{j,k}}^{j,k}]$, there are $K^{j,k}$ of them;
- ③ each node $v \in V$ corresponds to its aggregate differentiable function $\mathbf{agg}_v$.
- ④ The function $\mathbf{f} = \mathbf{f}_{|V|-1}$ is given as

$$\mathbf{f}_v(\mathbf{w}, \mathbf{x}) = \mathbf{agg}_v \left(\{ \langle \gamma^{j,k}, \mathbf{g}^{j,k} \rangle \circ \mathbf{f}_j(\mathbf{x}) \mid j \in \text{Adj}(v_k) \} \right), v \in \{1, \dots, |V|-1\}, \quad \mathbf{f}_0(\mathbf{x}) = \mathbf{x}. \quad (1)$$

It maps feature space $\mathbb{X}$ to label space $\mathbb{Y}$, given $\gamma^{j,k} \in [0, 1]^{K^{j,k}}$.

Def.

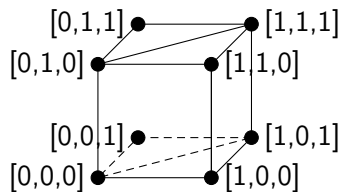
Call the graph (V, E) , the set of basic functions $\{\mathbf{g}^{j,k}, (j, k) \in E\}$ and the aggregate function $\{\mathbf{agg}_v, v \in V\}$ *the parametric family of models* $\mathfrak{F}$.

Statement

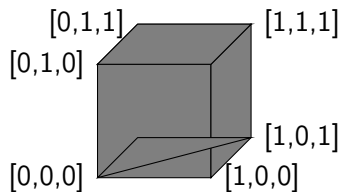
For any $\gamma^{j,k} \in [0, 1]^{K^{j,k}}$ the function $\mathbf{f} \in \mathfrak{F}$ is a model.

The restrictions to the structure parameters

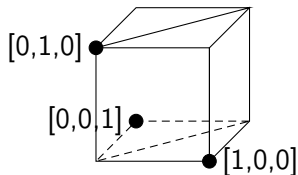
Examples for some structure parameter γ , $|\gamma| = 3$.



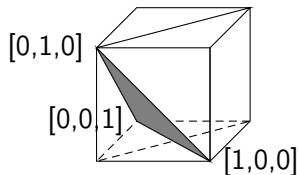
On vertices of the cube



Inside the cube



On vertices of the simplex

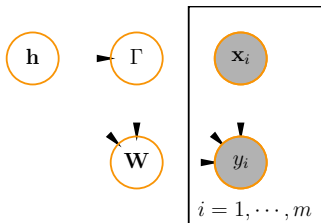


Inside the simplex

The prior distribution of parameters

Def.

Call *the prior* of the parameters $\mathbf{w}$ and the structure Γ of the model $\mathbf{f}$ the distribution $p(\mathbf{W}, \Gamma | \mathbf{h}, \mathbf{f}) : \mathbb{W} \times \Gamma \times \mathbb{H} \rightarrow \mathbb{R}^+$, where $\mathbb{W}$ is model parameter space, Γ is model structure space.



Def.

Call *the hyperparameters* $\mathbf{h} \in \mathbb{H}$ the parameters of distribution $p(\mathbf{w}, \Gamma | \mathbf{h}, \mathbf{f})$ (parameters of the distribution of parameters of the model $\mathbf{f}$).

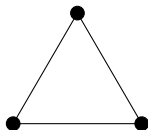
The model $\mathbf{f}$ is defined by the following variables:

- **The parameters** $\mathbf{w} \in \mathbb{W}$ narrows the model $\mathbf{f}$ to some function.
- **The structure parameters** $\Gamma = \{\gamma^{j,k}\}_{(j,k) \in E} \in \Gamma$ define a superposition to form the model $\mathbf{f}$.
- **Hyperparameters** $\mathbf{h} \in \mathbb{H}$ define the distribution of the parameters and structure parameters of the model.
- **Metaparameters** $\lambda \in \Lambda$ defined the optimization algorithm to select a model.

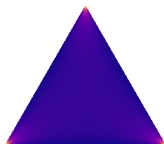
Prior distribution of the model structure

Each vector in the simplex defines some superposition.

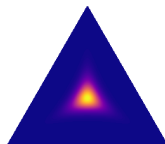
Gumbel-softmax: $\boldsymbol{\Gamma} \sim \text{GS}(\mathbf{s}, \lambda_{\text{temp}})$



$$\lambda_{\text{temp}} \rightarrow 0$$

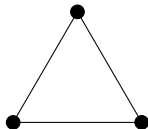


$$\lambda_{\text{temp}} = 0.995$$

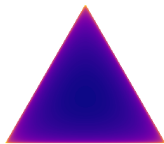


$$\lambda_{\text{temp}} = 5.0$$

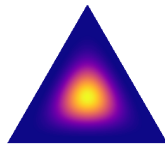
Dirichlet: $\boldsymbol{\Gamma} \sim \text{Dir}(\mathbf{s}, \lambda_{\text{temp}})$



$$\lambda_{\text{temp}} \rightarrow 0$$



$$\lambda_{\text{temp}} = 0.995$$

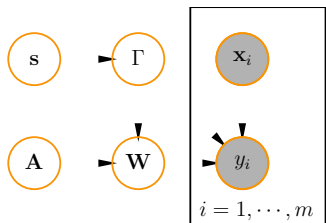


$$\lambda_{\text{temp}} = 5.0$$

Bayesian model selection

Basic model:

- parameters of the model
 $\mathbf{w} \sim \mathcal{N}(0, \alpha^{-1})$,
- hyperparameters
 $\mathbf{h} = [\alpha]$.



New model:

- parameters of the model
 $\mathbf{w}_r^{j,k} \sim \mathcal{N}(0, \gamma_r^{j,k} (\mathbf{A}_r^{j,k})^{-1})$, $\mathbf{A}_r^{j,k}$ is the diagonal matrix of the basic functions parameters $\mathbf{g}_r^{j,k}$,
 $(\mathbf{A}_r^{j,k})^{-1} \sim \text{inv-gamma}(\lambda_1, \lambda_2)$,
- structure parameters of the model
 $\Gamma = \{\gamma^{j,k}, (j, k) \in E\}$,
 $\gamma^{j,k} \sim \text{GS}(\mathbf{s}^{j,k}, \lambda_{\text{temp}})$,
- hyperparameters $\mathbf{h} = [\text{diag}(\mathbf{A}), \mathbf{s}]$,
- metaparameters $\lambda_1, \lambda_2, \lambda_{\text{temp}}$.

The model evidence as statistical complexity

The **statistical complexity** of the model f according to the Minimum Description Length principle:

$$\text{MDL}(\mathbf{y}, f) = -\log p(\mathbf{h}|f) - \log p(\hat{\mathbf{w}}|\mathbf{h}, f) - \log (p(\mathbf{y}|\mathbf{X}, \hat{\mathbf{w}}, f)\delta\mathfrak{D}),$$

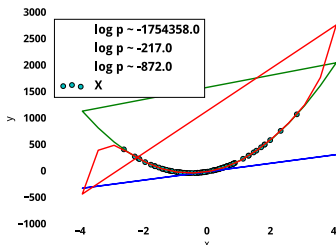
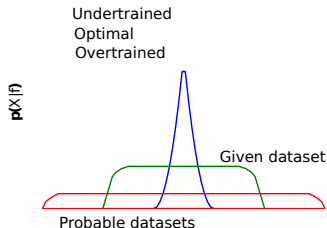
where $\delta\mathfrak{D}$ is the admissible precision of transferring the information on the dataset $\mathfrak{D}$.

The parameters $\mathbf{w}$ are optimized according to their **posterior distribution**:

$$L = \log p(\mathbf{w}|\mathbf{X}, \mathbf{y}, \mathbf{h}, f) \propto \log p(\mathbf{y}|\mathbf{X}, \mathbf{w}, \mathbf{h}, f) + \log p(\mathbf{w}|\mathbf{h}, f).$$

Hyperparameters are optimized according to their **prior distribution**:

$$Q = \log p(f|\mathbf{X}, \mathbf{y}) \propto \log p(\mathbf{h}|f) + \log \int p(\mathbf{y}|\mathbf{X}, \mathbf{w}, f)p(\mathbf{w}|\mathbf{h}, f)d\mathbf{w}.$$



The variational evidence lower bound

In general case the evidence could not be integrated analytically.

The model evidence:

$$p(\mathbf{y}|\mathbf{X}, \lambda_{\text{temp}}, \mathbf{f}) = \iint_{\mathbf{w}, \Gamma} p(\mathbf{y}|\mathbf{X}, \mathbf{w}, \Gamma, \mathbf{f}) p(\mathbf{w}, \Gamma|\mathbf{h}, \lambda_{\text{temp}}, \mathbf{f}) d\mathbf{w} d\Gamma.$$

Def.

Call the *variational parameters* of the model $\theta \in \mathbb{R}^u$ the parameters of distribution q that approximate the prior of the parameters and the structure $p(\mathbf{w}, \Gamma|\mathbf{X}, \mathbf{y}, \mathbf{h}, \mathbf{f}, \lambda_{\text{temp}})$:

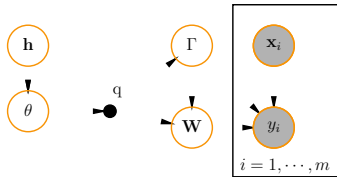
$$q \approx \iint_{\mathbf{w}', \Gamma'} p(\mathbf{y}|\mathbf{X}, \mathbf{w}', \Gamma', \mathbf{f}) p(\mathbf{w}', \Gamma'|\mathbf{h}, \lambda_{\text{temp}}, \mathbf{f}) d\mathbf{w}' d\Gamma'$$

Obtain the lower bound estimation $\log \hat{p}(\mathbf{y}|\mathbf{X}, \lambda_{\text{temp}}, \mathbf{f})$ of the integral

$$\log p(\mathbf{y}|\mathbf{X}, \lambda_{\text{temp}}, \mathbf{f}) \geq E_q \log p(\mathbf{y}|\mathbf{X}, \mathbf{w}, \Gamma, \mathbf{f}) - D_{\text{KL}}(q(\mathbf{w}, \Gamma) || p(\mathbf{w}, \Gamma|\mathbf{h}, \lambda_{\text{temp}}, \mathbf{f})).$$

It equals to the integral of the evidence when

$$D_{\text{KL}}(q(\mathbf{w}, \Gamma) || p(\mathbf{w}, \Gamma|\mathbf{y}, \mathbf{X}, \lambda_{\text{temp}}, \mathbf{f})) = 0.$$



The problem of model selection

Fixate the variational distribution $q = q_{\mathbf{w}}q_{\mathbf{r}}$ and its parameters θ , that approximate the posterior $p(\mathbf{w}, \mathbf{r} | \mathbf{X}, \mathbf{y}, \mathbf{h}, \mathbf{f})$ of the model parameters and the structure.

Def.

Call the *loss function* $L(\theta | \mathbf{h}, \mathbf{X}, \mathbf{y}, \mathbf{f})$ the differentiable function, a model quality on some training dataset given the parameters θ on the distribution q .

Call the *validation function* $Q(\mathbf{h} | \theta, \mathbf{X}, \mathbf{y}, \mathbf{f})$ the differentiable function, a model quality, given θ indirectly

Call *the problem of model selection* $\mathbf{f}$ the two-level optimization problem:

$$\mathbf{h}^* = \arg \max_{\mathbf{h} \in \mathbb{H}} Q(\mathbf{h} | \theta^*, \mathbf{X}, \mathbf{y}, \mathbf{f}),$$

where θ^* is the solution of the optimization problem

$$\theta^* = \arg \max_{\theta \in \mathbb{R}^u} L(\theta | \mathbf{h}, \mathbf{X}, \mathbf{y}, \mathbf{f}).$$